

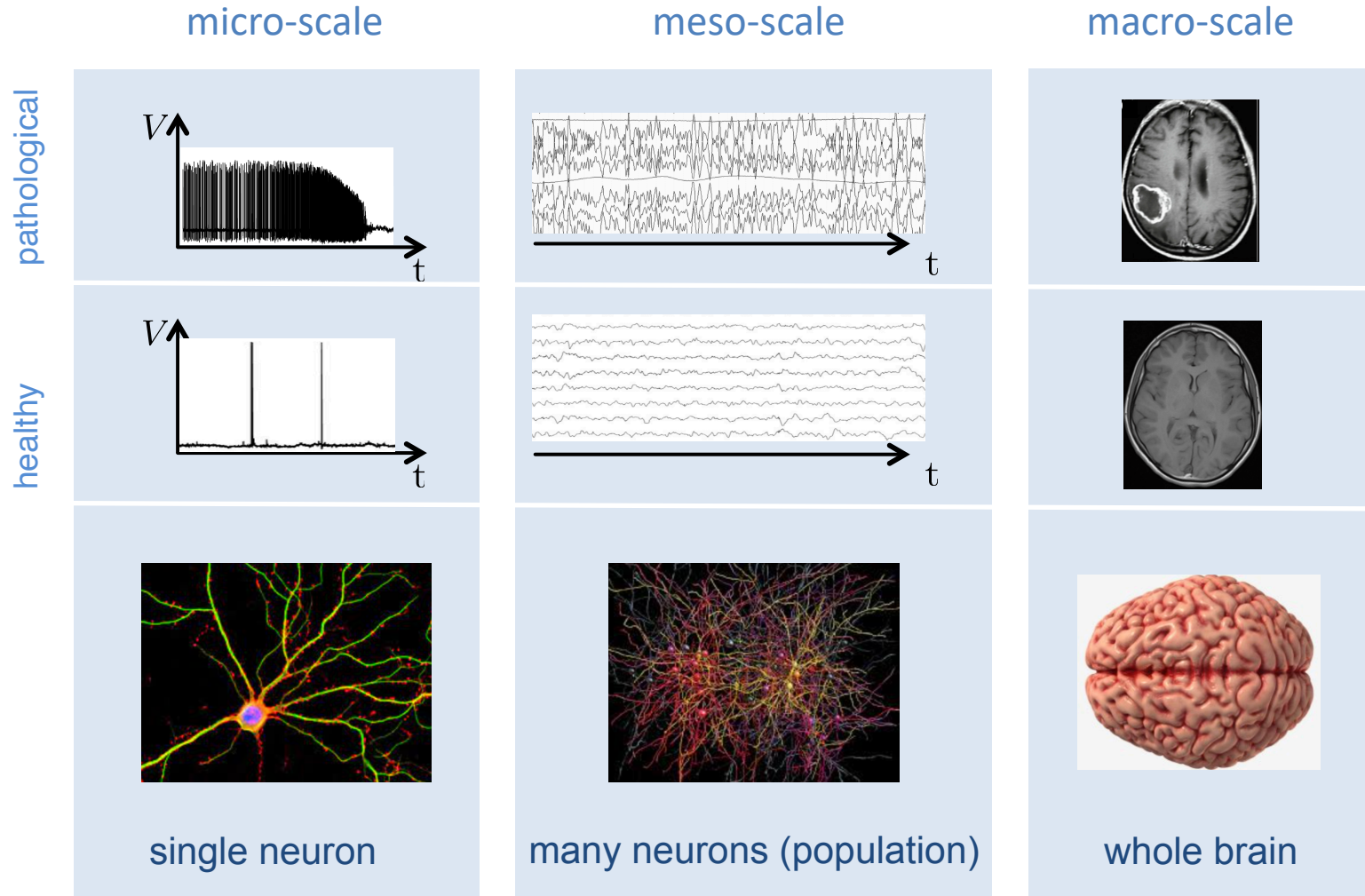
Summary:

Neural fields for modeling population dynamics

Emre Baspinar

Inria Montpellier / MathNeuro Team

Computational modeling of neural populations



Chever et al.
J. of Clinical Inv. 2021

Lemaire et al.
PLOS C. Bio. 2021

Tschernatsch et al.
Int. Care Med. 2014

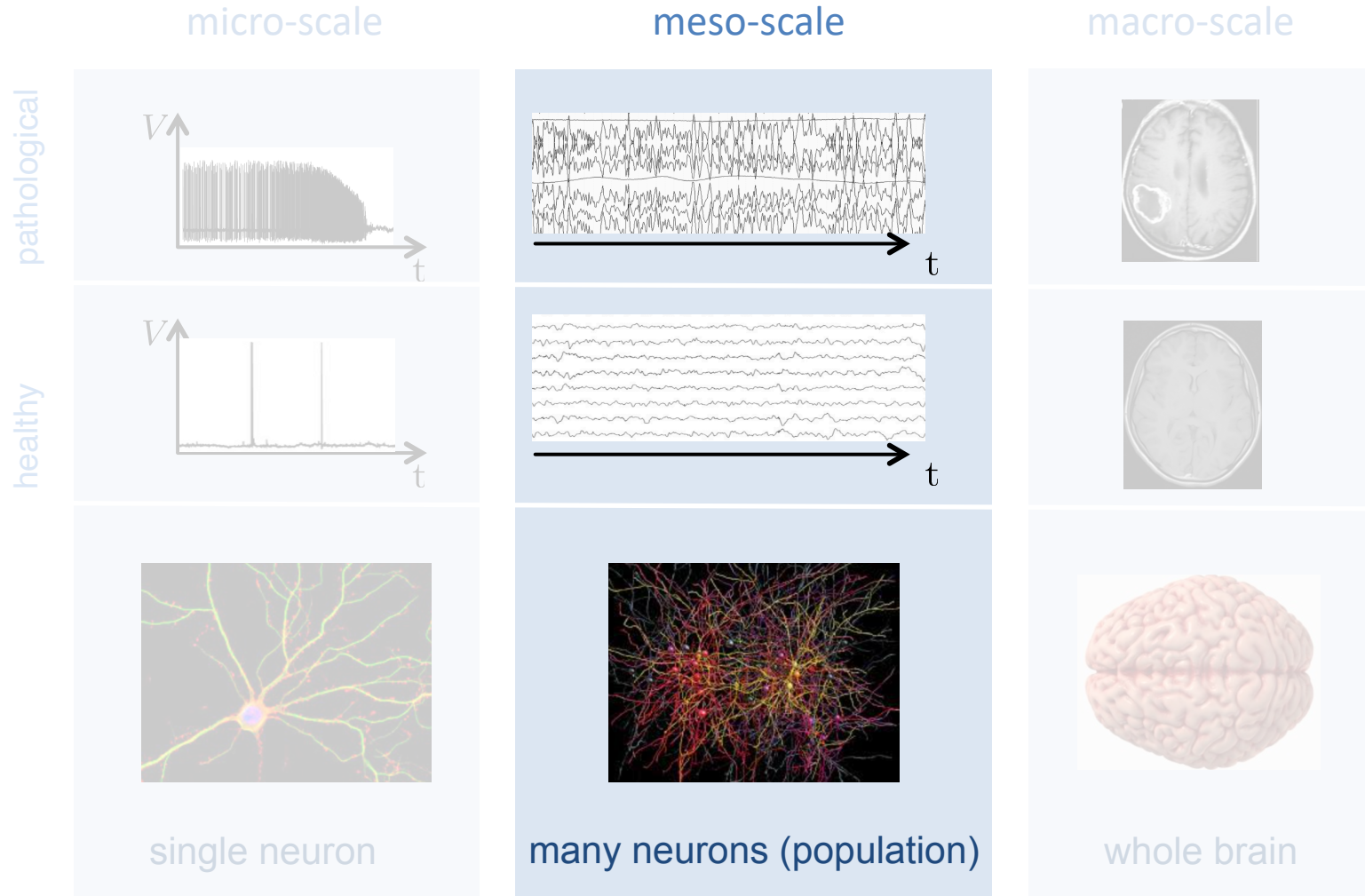
Plouin et al.
H. of Cl. Neuro. 2013

Liu et al.
Tsinghua Sci. Tech. 2014

example: micro-electrode

example: EEG

example: MRI



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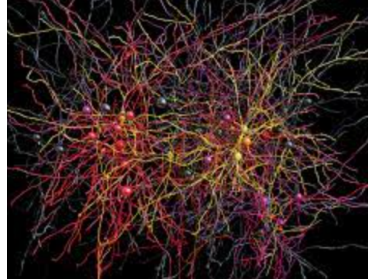
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example: micro-electrode

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Modeling neural populations



Plouin et al.
H. of Cl. Neuro. 2013

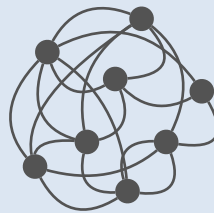
Networks

$$\frac{du_1}{dt} = f_1(u_1, \dots, u_n, t)$$

$$\frac{du_2}{dt} = f_2(u_2, \dots, u_n, t)$$

⋮

$$\frac{du_n}{dt} = f_n(u_1, \dots, u_n, t)$$



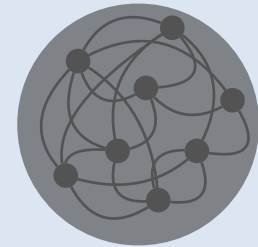
+ biophysically detailed
+ precise dynamics

- high dimensional
- difficult to manipulate
- computationally heavy

Mean-field approximation

$$\frac{dU}{dt} = F(U, t)$$

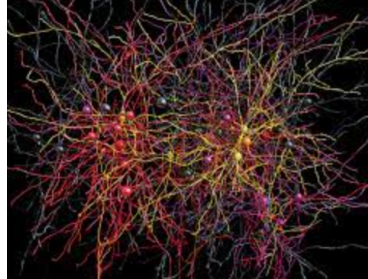
averaged dynamics



+ low dimensional
+ simple
+ computationally light

- not detailed
- approximate dynamics

Modeling neural populations



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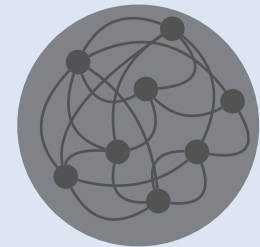
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Mean-field approximation

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averaged dynamics



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Mean-field models

Type	Domain	Neural activity	Examples
Localized Wilson-Cowan	Time	Oscillations Multi-stability Temporal dynamics	Wilson-Cowan 1972
Neural mass models	Time	Oscillations Multi-stability Temporal dynamics	Zetterberg-Jansen 1990s Jansen—Rit 1995 Wendling 2000s
Neural field models	Time and space	Travelling waves Pattern formation (bumps) Seizure propagation	Wilson-Cowan 1973 Amari 1977
Biophysical (derived) mean-field	Time Time and space	Asynch. irregular states Up-down states Balanced exc-inh regimes	LIF mean-field: - Brunel-Hakim 1999 AdEx mean-field: - El Boustani-Destexhe 2009 - Zerlaut et al. 2018 - di Volo et al. 2021

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Localized Wilson-Cowan model '72 (2-dim ODE)

Spatially-extended Wilson-Cowan model '73 (2-dim IDE)

Amari model '77 (1-dim IDE)

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Localized Wilson-Cowan model

Wilson-Cowan, Biophysical J, 1972

Modeling assumptions

1. Neurons are in close proximity:
 - spatial differences neglected
 - only temporal dynamics considered
2. Neural activity can be represented in terms of proportion of activated cells:
 - activity level represented by an instantaneous normalized quantity
3. Neural processes are dependent on the interaction of excitatory and inhibitory neurons:
 - explicitly represented excitatory and inhibitory population activity

Localized Wilson-Cowan model

Wilson-Cowan, Biophysical J, 1972

$$\frac{dE}{dt} = -E + (1 - E) S_e \left(w_{ee} E - w_{ei} I + P \right)$$

$$\frac{dI}{dt} = -I + (1 - I) S_i \left(w_{ie} E - w_{ii} I + Q \right)$$

$$E, I : [0, \infty) \rightarrow [0, 1]$$

proportion of activated excitatory or inhibitory neurons

$$P, Q : [0, \infty) \rightarrow \mathbb{R}$$

external stimuli

$$S_e, S_i : \mathbb{R} \rightarrow [0, 1]$$

population transfer functions

$$w_{ee}, w_{ei}, w_{ie}, w_{ii} > 0$$

connectivity weights

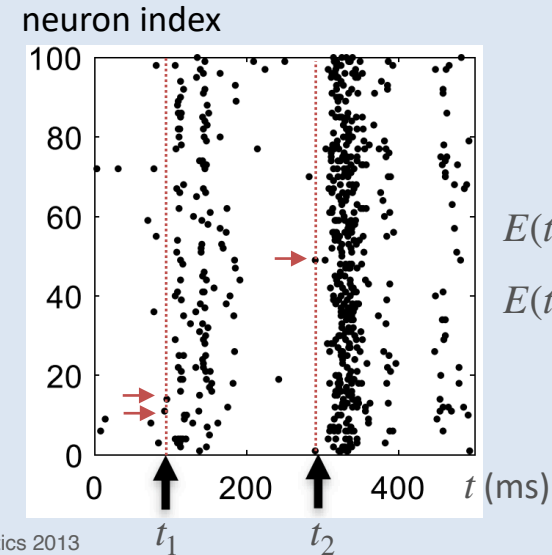
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Example:



$E, I : [0, \infty) \rightarrow [0, 1]$

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Localized Wilson-Cowan model

Wilson-Cowan, Biophysical J, 1972

$$\frac{dE}{dt} = \underbrace{-E}_{\text{negative feedback}} + \underbrace{(1-E)}_{\text{proportion of nonrefractory neurons}} \underbrace{S_e(w_{ee}E - w_{ei}I + P)}_{\text{synaptic input}}$$

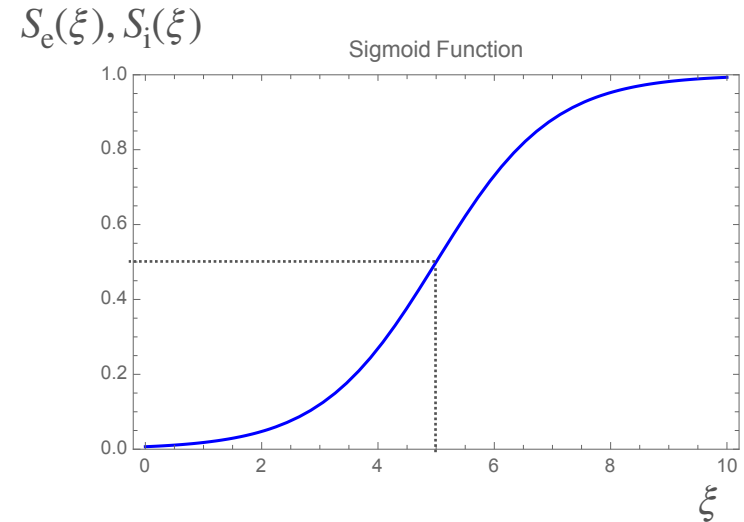
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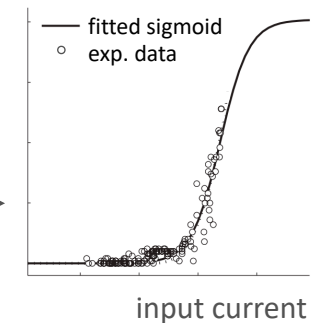


Why sigmoid?

- Low excitation activates no neuron → sigmoid min level
- High excitation activates all neurons → sigmoid max level
- Experimental evidence →

Fischer et al.
PLOS One, 2009

output firing rate



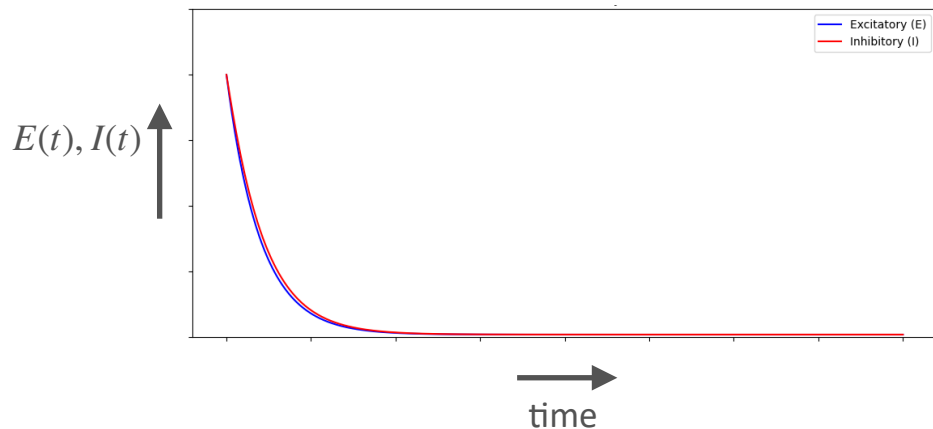
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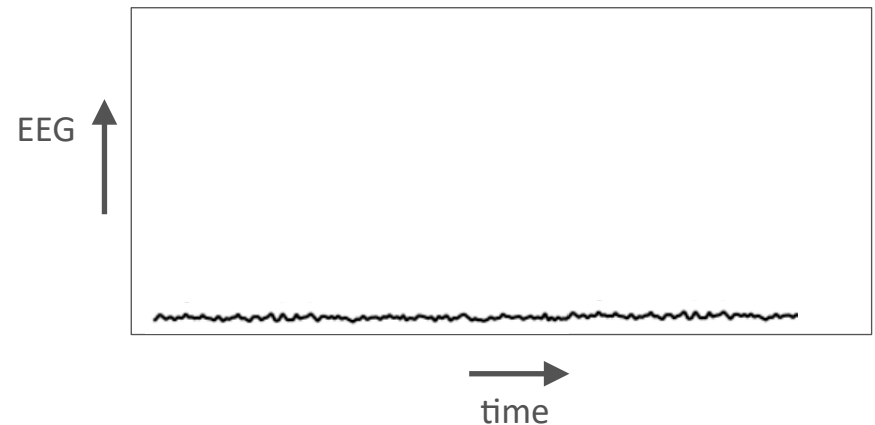
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Simulation



Experiment



adapted from
<https://frederic-vw.github.io/eeg-patterns/>

Commonly observed when there is no sensorial stimulus,
e.g., in task-free experiments

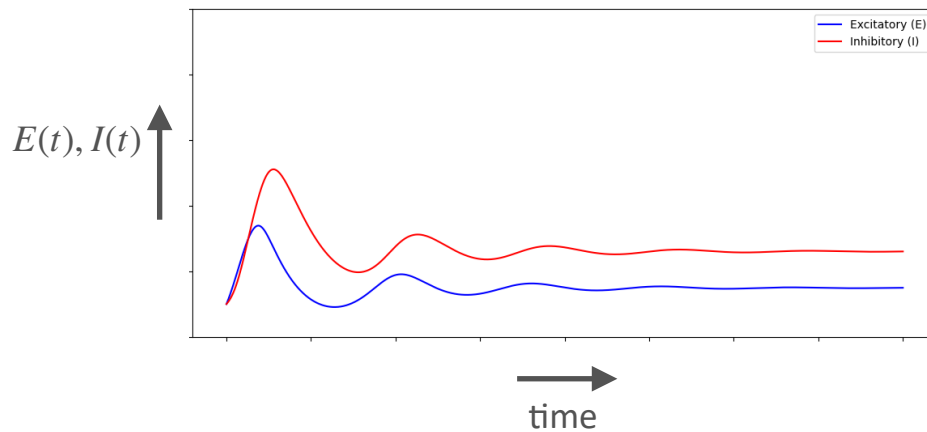
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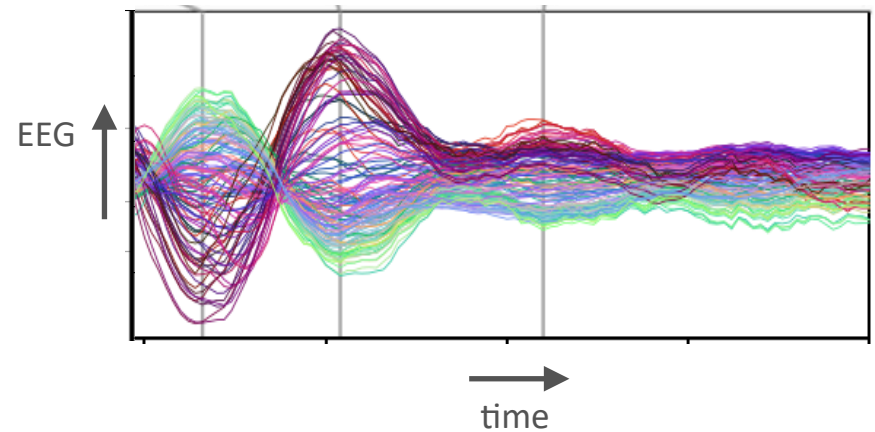
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Simulation



Experiment



Pascucci et al., Sci Data, 2022

Commonly observed in stimulus-evoked activity

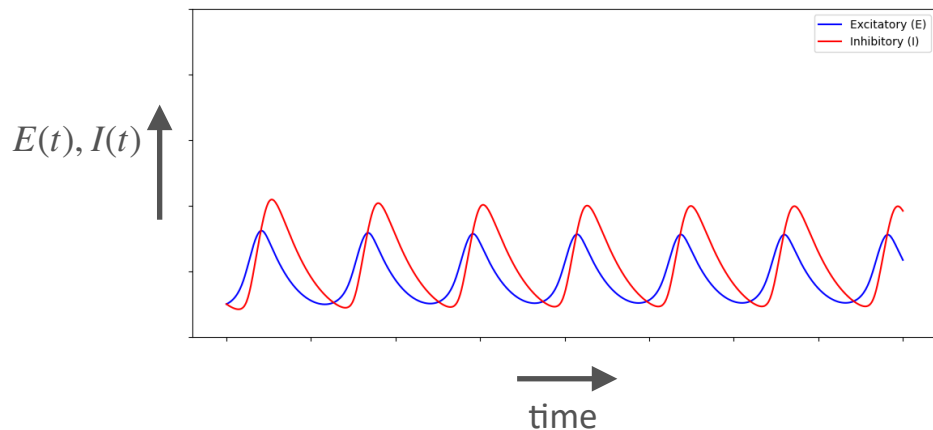
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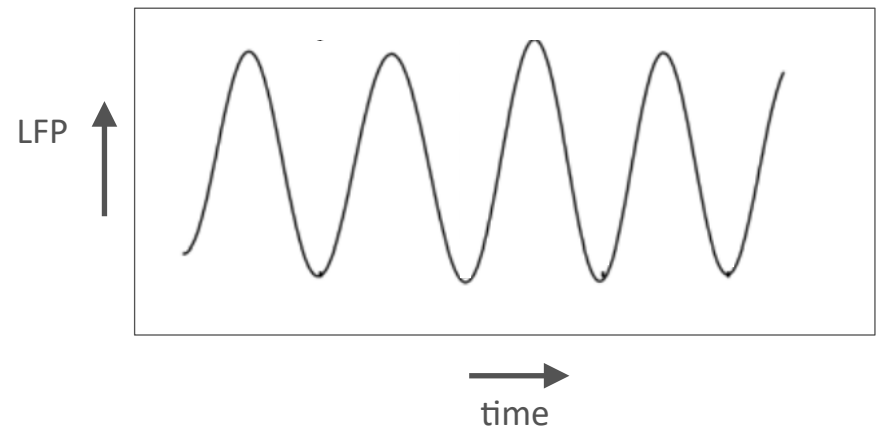
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Simulation



Experiment



Scheffzük et al., PLOS One, 2011

Commonly observed in cognitive tasks, or in different brain states
e.g., in REM sleep

Localized Wilson-Cowan model '72 (2-dim ODE)

Spatially-extended Wilson-Cowan model '73 (2-dim IDE)

Amari model '77 (1-dim IDE)

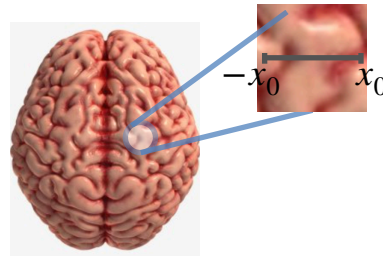
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Spatially-extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973



$$[-x_0, x_0] \subset \mathbb{R}$$

Localized Wilson-Cowan

$$\frac{dE}{dt}(t) = -E(t) + \left(1 - E(t)\right) S_e\left(w_{ee} E(t) - w_{ei} I(t) + P(t)\right)$$

$$\frac{dI}{dt}(t) = -I(t) + \left(1 - I(t)\right) S_i\left(w_{ie} E(t) - w_{ii} I(t) + Q(t)\right)$$

Spatially-extended Wilson-Cowan

$$\frac{\partial E}{\partial t}(x, t) = -E(x, t) + \left(1 - E(x, t)\right) S_e\left((E \circledast \kappa_{ee})(x, t) - (I \circledast \kappa_{ei})(x, t) + P(x, t)\right)$$

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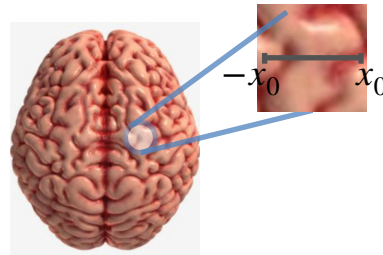


proximity
assumption fails



Spatially-extended Wilson-Cowan model

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Localized Wilson-Cowan

connectivity weights

$$\frac{dE}{dt}(t) = -E(t) + \left(1 - E(t)\right) S_e \left(\overset{\downarrow}{w_{ee}} E(t) - \overset{\downarrow}{w_{ei}} I(t) + P(t) \right)$$

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connectivity kernels

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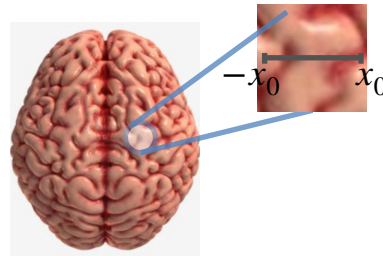


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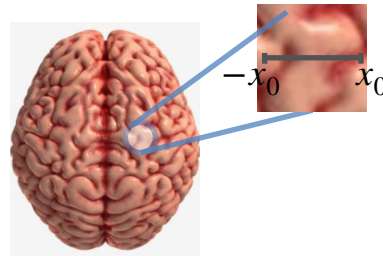


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proximity
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Spatially-extended Wilson-Cowan model

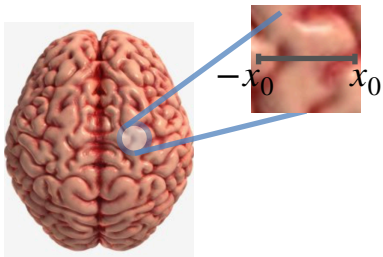
Wilson-Cowan, Kybernetik, 1973

Spatially-extended Wilson-Cowan

connectivity kernels

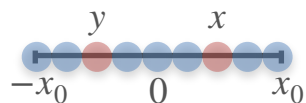
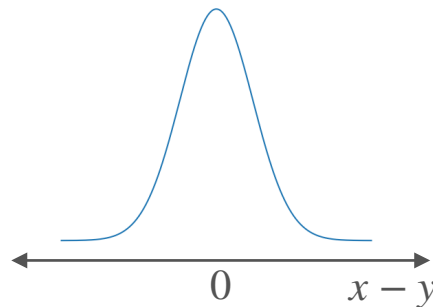
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Connectivity kernel

$$\kappa_{\square}(x, y) := e^{-\frac{(x-y)^2}{\sigma_{\square}}}$$



- isotropic: **only** distance dependent
- homogeneous: no change in time
- symmetric: $\kappa_{\square}(x, y) = \kappa_{\square}(y, x)$

Spatially-extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

Spatially-extended Wilson-Cowan

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- active transients

- spatially localized limit cycles

+

temporal patterns

- travelling waves

Let's have a look at the simulations!

Localized Wilson-Cowan model '72 (2-dim ODE)

Spatially-extended Wilson-Cowan model '73 (2-dim IDE)

Amari model '77 (1-dim IDE)

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Amari model

$$\frac{\partial U}{\partial t}(x, t) = -U(x, t) + \left(S(U) \circledast \kappa \right)(x, t) + P(x, t)$$

- U denotes the average membrane potential of a population.
- There is no explicit representation of refractoriness.
- Nonlinear transform of neural activity is before the convolution.
- Connectivity kernel encodes both excitation and inhibition.

Spatially-extended Wilson-Cowan

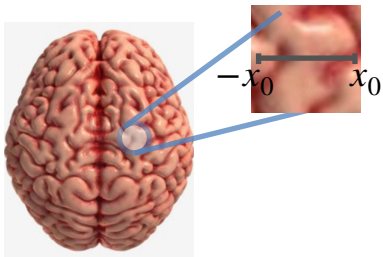
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- E, I denotes the proportions of active neurons in exc. and inh. populations.
- Refractoriness is expressed explicitly.
- Nonlinear transform of neural activity is after the convolution.
- Connectivity kernels for excitation and inhibition are separate.

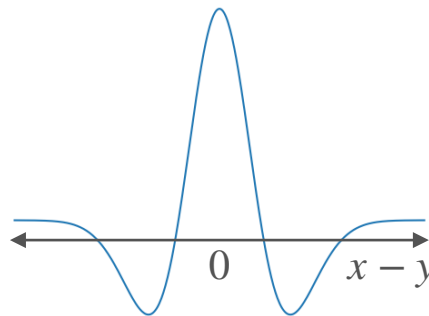
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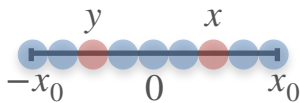


Connectivity kernel

$$\kappa(x, y) := \left(1 - \frac{(x - y)^2}{\sigma^2} \right) e^{-\frac{(x - y)^2}{2\sigma^2}}$$

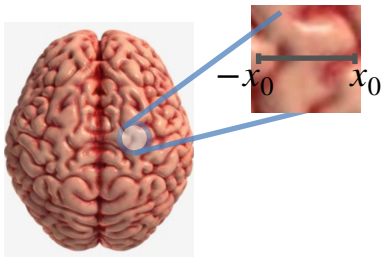


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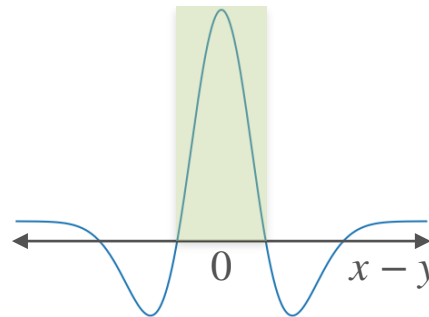
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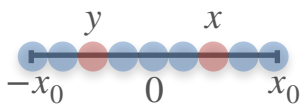
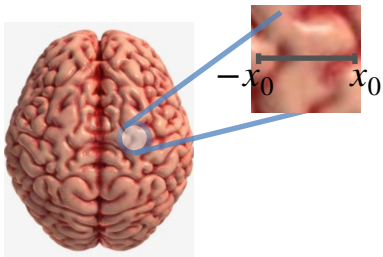


Short-range excitation

- isotropic: **only** distance dependent
- homogeneous: no change in time
- symmetric: $\kappa(x, y) = \kappa(y, x)$

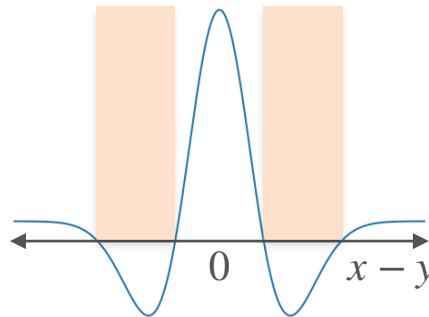
Amari model

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Connectivity kernel

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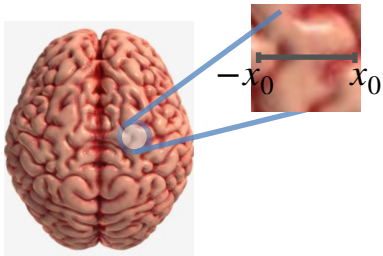
Long-range inhibition

- isotropic: **only** distance dependent
- homogeneous: no change in time
- symmetric: $\kappa(x, y) = \kappa(y, x)$

Variants of Amari model

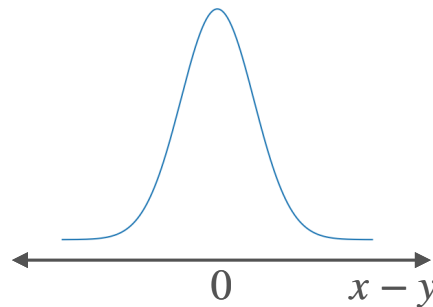
$$\frac{\partial U_e}{\partial t}(x, t) = -U_e(x, t) + \left(S(U_e) \circledast \kappa_{ee} \right)(x, t) - \left(S(U_i) \circledast \kappa_{ei} \right)(x, t) + P(x, t)$$

$$\frac{\partial U_i}{\partial t}(x, t) = -U_i(x, t) + \left(S(U_e) \circledast \kappa_{ie} \right)(x, t) - \left(S(U_i) \circledast \kappa_{ii} \right)(x, t) + Q(x, t)$$

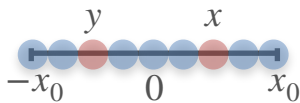


Connectivity kernel

$$\kappa_{\square}(x, y) := e^{-\frac{(x-y)^2}{\sigma_{\square}}}$$



- isotropic: **only** distance dependent
- homogeneous: no change in time
- symmetric: $\kappa_{\square}(x, y) = \kappa_{\square}(y, x)$



Variants of Amari model

$$\frac{\partial U_e}{\partial t}(x, t) = -U_e(x, t) + \left(S(U_e) \circledast \kappa_{ee} \right)(x, t) - \left(S(U_i) \circledast \kappa_{ei} \right)(x, t) + P(x, t)$$

$$\frac{\partial U_i}{\partial t}(x, t) = -U_i(x, t) + \left(S(U_e) \circledast \kappa_{ie} \right)(x, t) - \left(S(U_i) \circledast \kappa_{ii} \right)(x, t) + Q(x, t)$$

- active transients

- spatially localized limit cycles

+

temporal patterns

- travelling waves

Neuron fields provide coarse-grained approximations of neural populations dynamics

Localized Wilson-Cowan model '72 (2-dim ODE)

- Neurons in close proximity
- No space dependency
- Only temporal dynamics

Spatially-extended Wilson-Cowan model '73 (2-dim IDE)

- Neurons distributed in space
- Space dependency
- Connectivity depending on the distance between the interacting neurons

Amari model '77 (1-dim IDE)

- Neurons distributed in space
- Space dependency
- Connectivity depending on the distance between the interacting neurons
- Single population with excitation-inhibition encoded together in the connectivity

Summary:

Neurogeometry

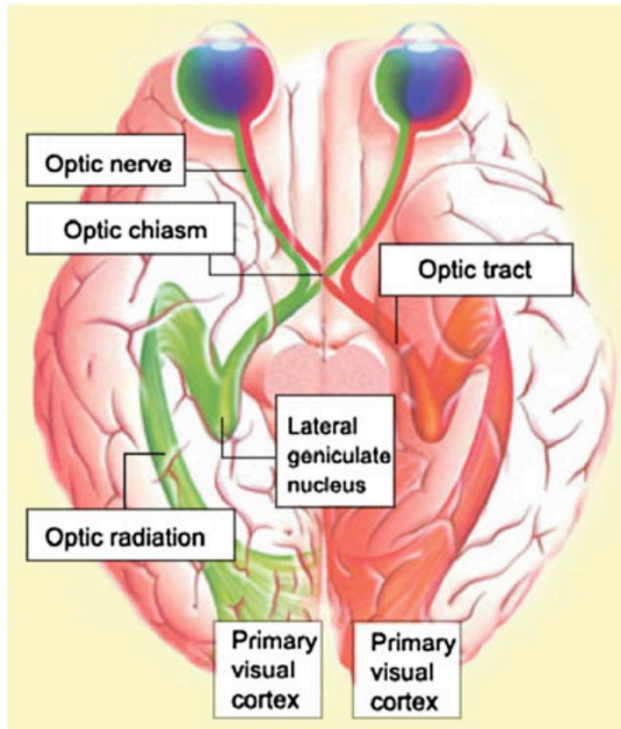
Emre Baspinar
Inria Montpellier / MathNeuro Team

Computational modeling of neural populations

Retino-geniculo-cortical pathway

Focus:

- Human, primates, cats (not all mammals, e. g., not mice!)
- First step of visual perception in the cortex — Primary visual cortex



Retina



Optic nerve

Optic chiasm: cross-point that organizes optic nerves properly for coordinated binocular vision



Optic tract

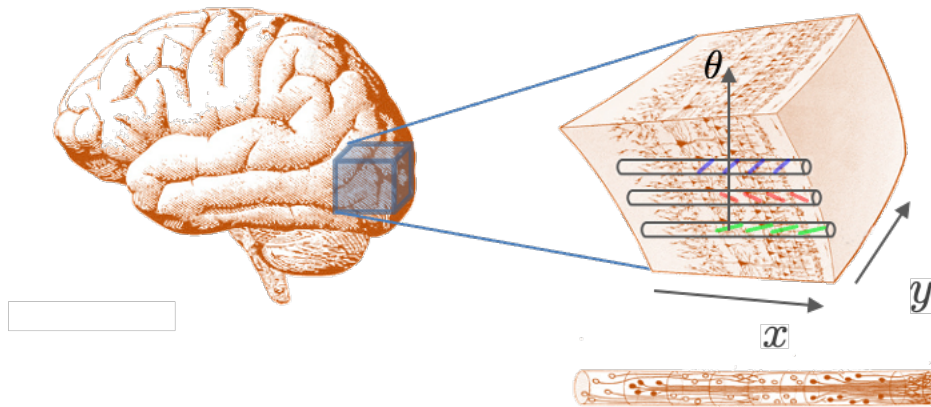
Lateral geniculate nucleus (LGN): thalamic relay transmitting visual information to the primary visual cortex



Optic radiation

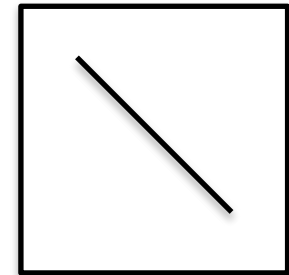
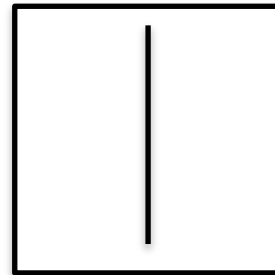
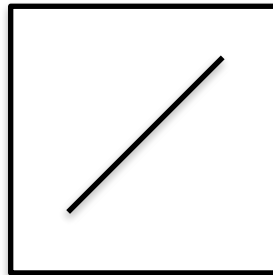
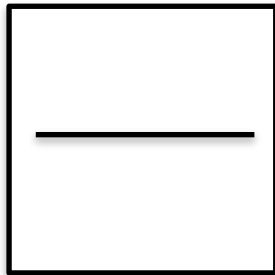
Primary visual cortex

Primary visual cortex (V1)

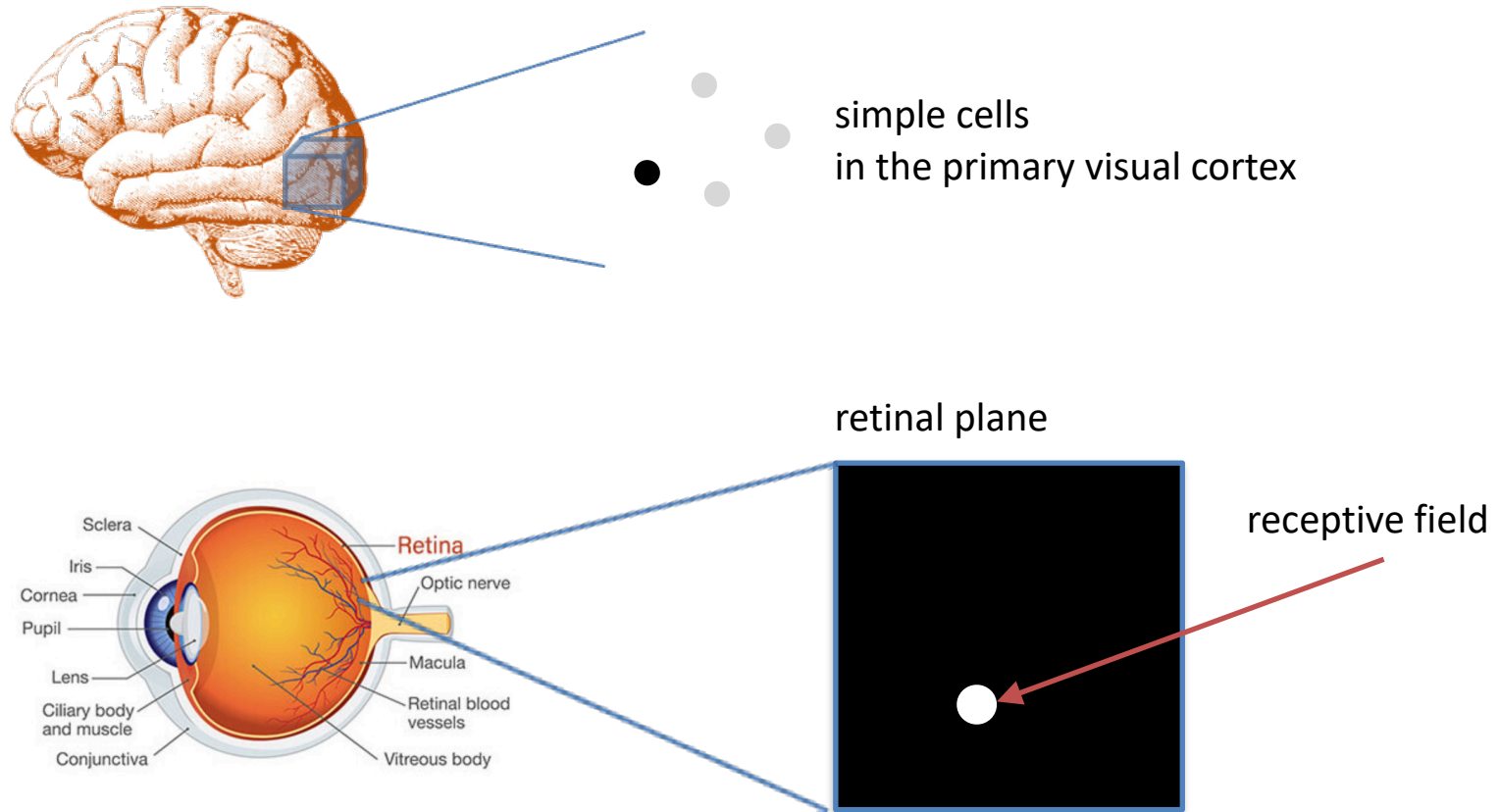


Simple cells

- Located in the occipital lobe (backside)
- Responsible for first step processing of the visual information transmitted from retina
- Sensitive to visual features of the input, **in particular to orientation**



Simple cells, receptive fields, receptive profiles

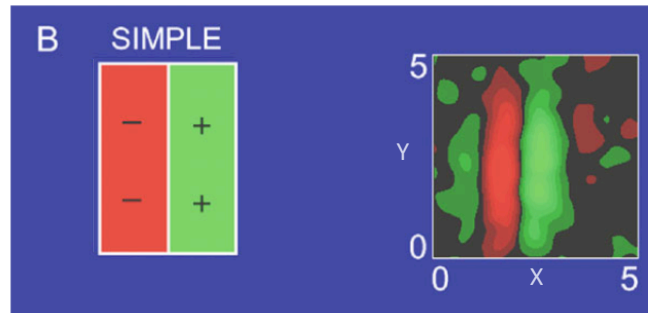


Receptive field:

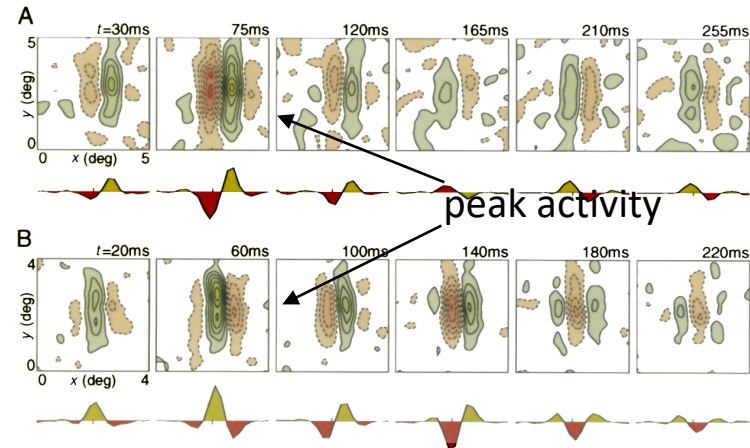
- Retinal zone to which the simple cell is connected through retino-geniculo-cortical pathways.
- Simple cell produces spikes (gives response) when its receptive field is stimulated.

Modeling simple cell receptive profiles

Receptive profile (experimental)

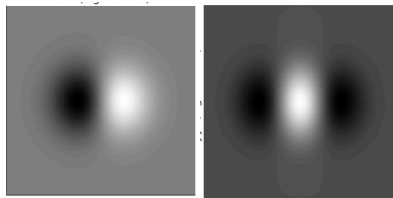


Petitot, Neurogéométrie de la vision, 2008

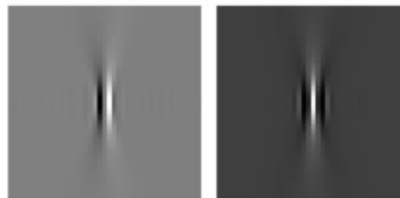


DeAngelis et al., Trends Neurosci., 1995

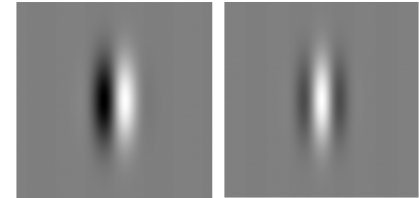
Receptive profile (mathematical model)



Gaussian derivatives



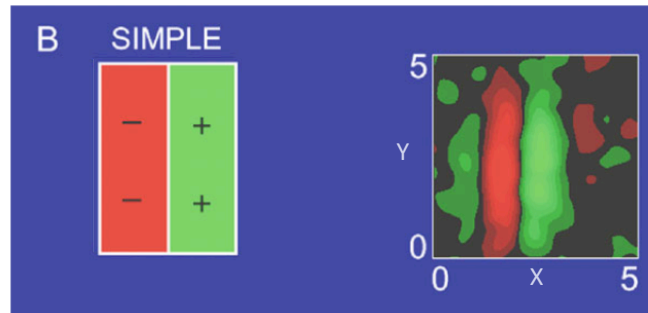
Gabor kernels



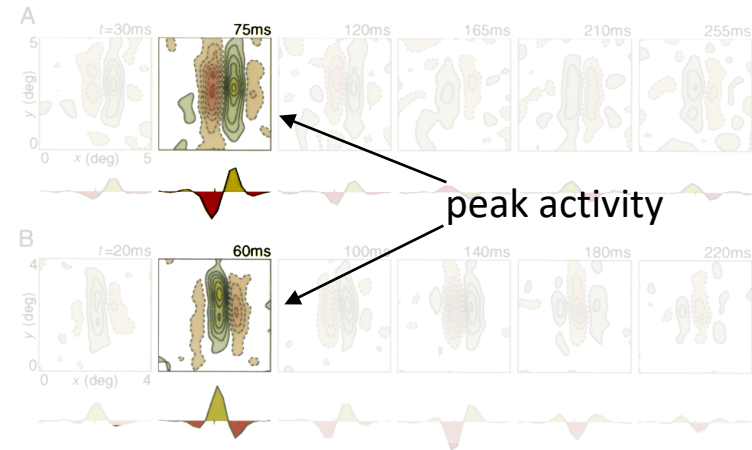
Gabor functions

Modeling simple cell receptive profiles

Receptive profile (experimental)

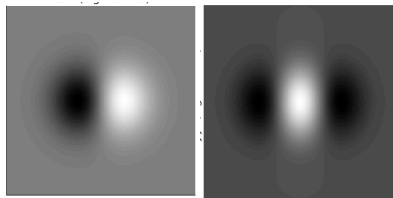


Petitot, Neurogéométrie de la vision, 2008

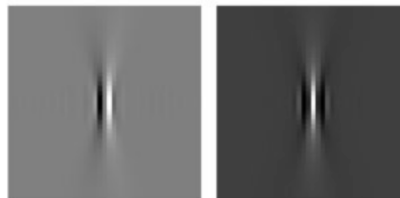


DeAngelis et al., Trends Neurosci., 1995

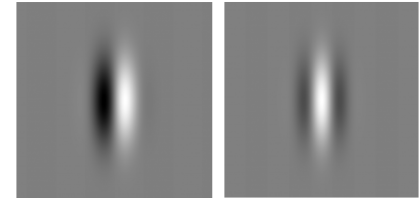
Receptive profile (mathematical model)



Gaussian derivatives



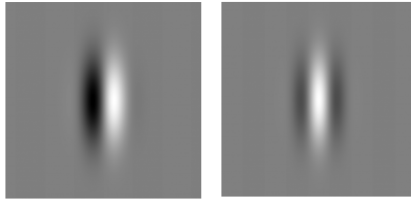
Gabor kernels



Gabor functions

Modeling simple cell receptive profiles

odd (sin)
component even (cos)
component



Gabor functions

$$(\exp(i x) = \cos x + i \sin x)$$



$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i \frac{\eta}{\sigma}\right)$$



Normalization
constant



Localizing Gaussian
window



Wave content

$$\begin{aligned}\xi &= (x - x_0) \cos \theta_0 + (y - y_0) \sin \theta_0 \\ \eta &= -(x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0\end{aligned}$$

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \quad R_{\theta_0} = \begin{pmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}_{\text{translation}} + \underbrace{R_{\theta_0}}_{\text{rotation}} \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$



Let us write this in terms of a transfer operator:

$$(x, y) = A_{(x_0, y_0, \theta_0)}(\xi, \eta).$$

So we can write (ξ, η) as follows:

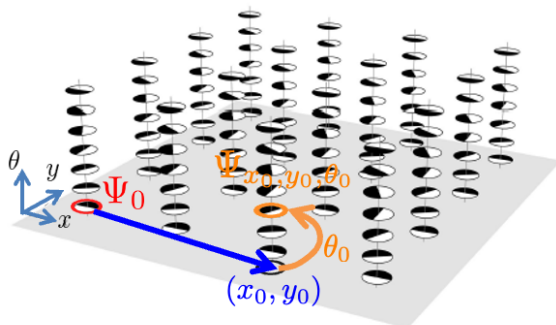
$$(\xi, \eta) = A_{(x_0, y_0, \theta_0)}^{-1}(x, y).$$

This provides the relation btwn $\Psi_{(0,0,0)}$ and $\Psi_{(x_0, y_0, \theta_0)}$:

$$\Psi_{(0,0,0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \exp\left(i \frac{y}{\sigma}\right)$$

$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \Psi_{(0,0,0)}(A_{(x_0, y_0, \theta_0)}^{-1}(x, y))$$

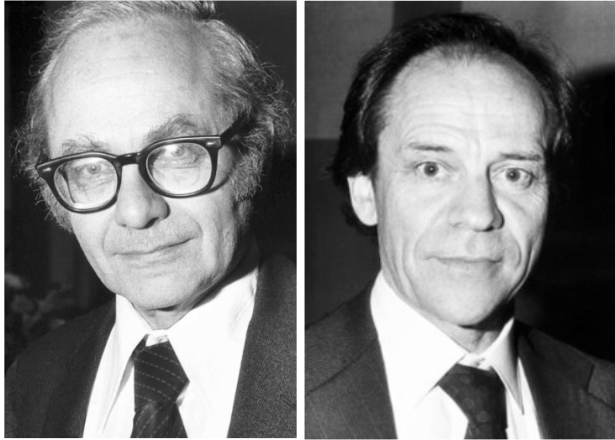
41



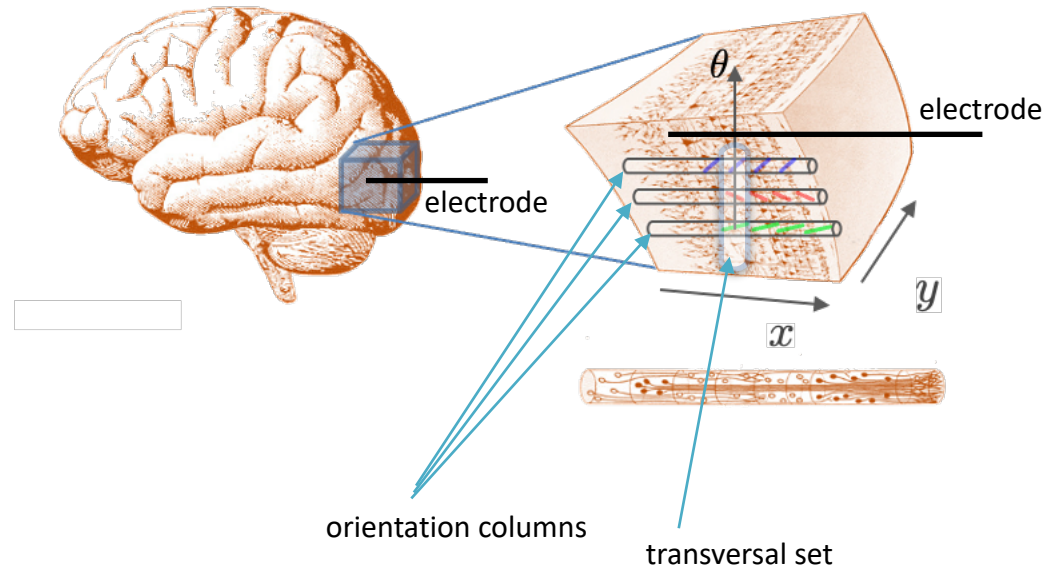
Set of receptive profiles

Functional architecture of the primary visual cortex

David H. Hubel & Torsten N. Wiesel

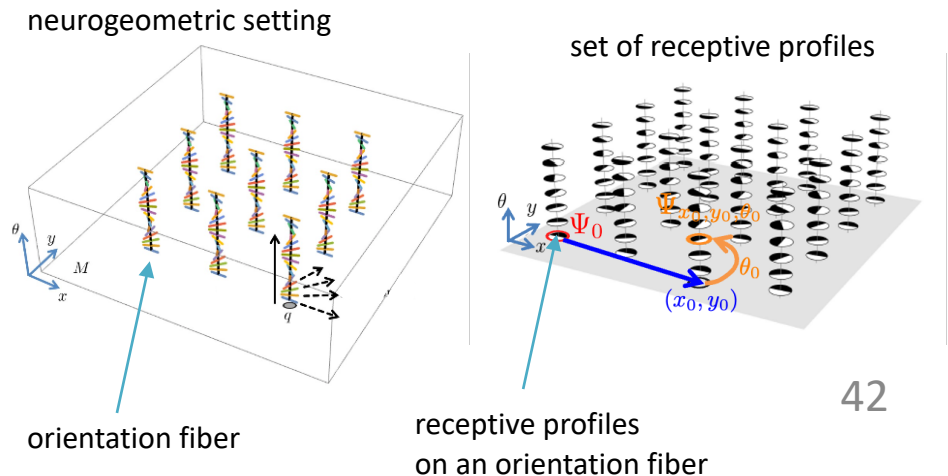


nobelprize.org



Columnar organization of simple cells

- Transversal sets are distributed in a periodic (spatial) fashion on the cortical surface.
- This is similar to the set of receptive profiles.
- This is the key point linking the columnar functional architecture of the primary visual cortex to the set of receptive profiles.



Citti-Sarti-Petitot model of the functional architecture

Model assumptions:

- No motion. 2D grayscale images are assumed as visual input.
 - Restricted to a single instant.
 - All kinds of movements (head, eye) are excluded, not only the movements in a visual scene.
 - Therefore, no dynamics are taken into account. It is a static model.
- Retino-cortical mapping is ignored (assumed to be identical mapping).
 - It maps the retinal coordinates to cortical coordinates.
 - Image plane, retinal plane and cortical plane are assumed to be identical.
- Receptive profiles do not change in time, they are static.
 - Neural response to a given visual input does not change.
- Neural responses are linear.
 - Response to a given visual input can be obtained via a linear filtering with Gabor functions (lifting).

Focuses only on orientation-selective behavior of the simple cells in V1!

Selectivity to other visual features (scale, spatial frequency etc.) is not taken into account!
(But extensions to other features exist in the modeling literature)

Citti-Sarti-Petitot model of the functional architecture

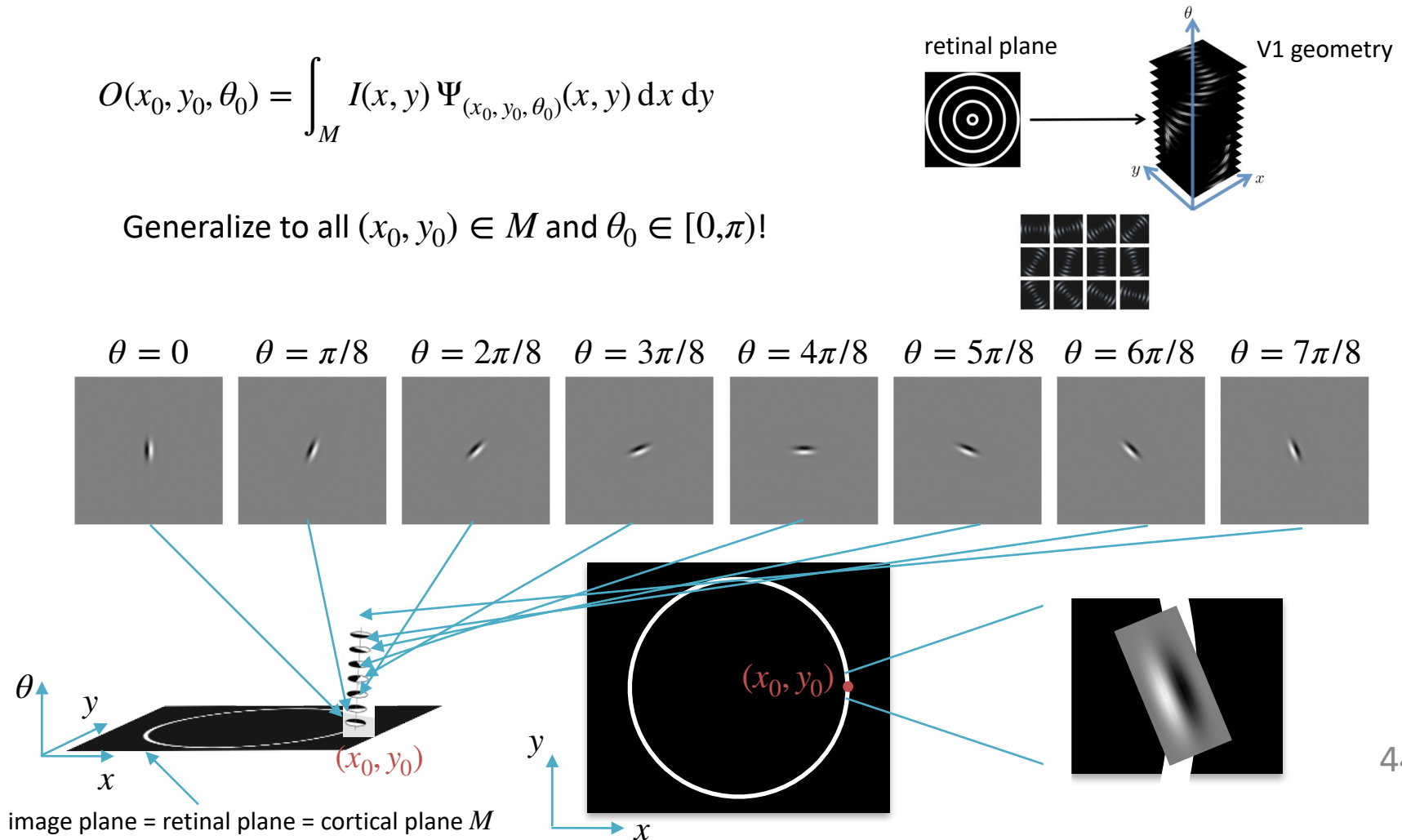
We restrict ourselves to orientation selectivity!

What do we mean by orientation selectivity?

Overlap of orientation of a simple cell receptive profile with the orientation of visual input.

$$O(x_0, y_0, \theta_0) = \int_M I(x, y) \Psi_{(x_0, y_0, \theta_0)}(x, y) dx dy$$

Generalize to all $(x_0, y_0) \in M$ and $\theta_0 \in [0, \pi)$!

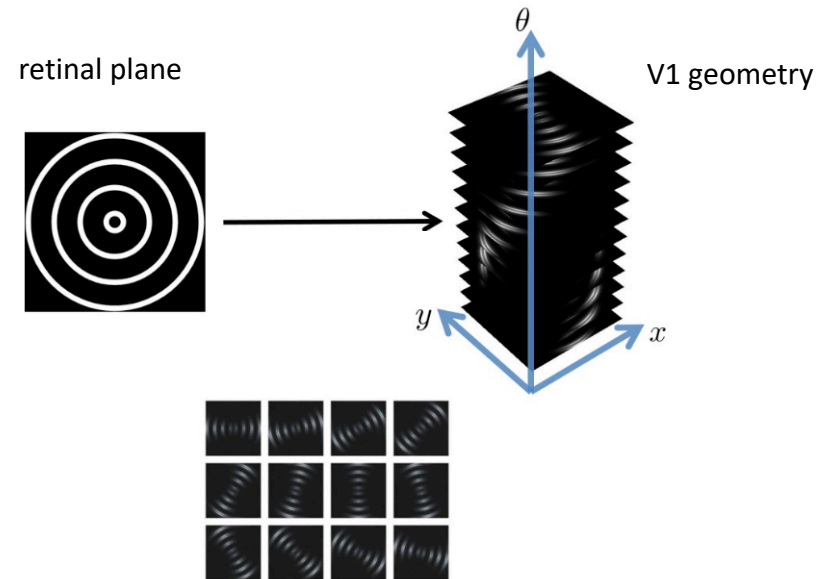


Citti-Sarti-Petitot model of the functional architecture

$$O(x_0, y_0, \theta_0) = \int_M I(x, y) \Psi_{(x_0, y_0, \theta_0)}(x, y) dx dy$$



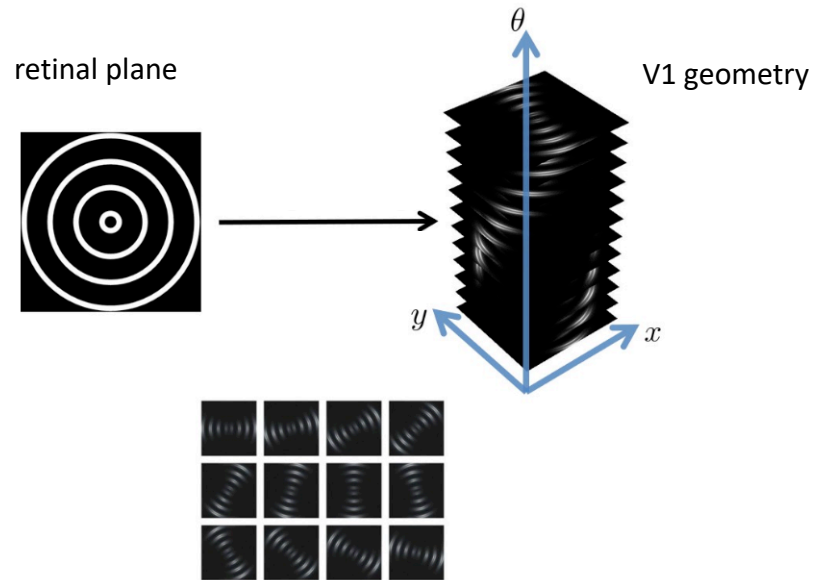
Output response of the simple cell selective to the position $(x_0, y_0) \in M$ and to the orientation $\theta_0 \in [0, \pi)$



- Output responses are obtained via filtering of the input image with Gabor functions: Gabor transform.
- Output response is a complex value since the wave content of $\Psi_{(x_0, y_0, \theta_0)}$ is a complex number.
- In general, we choose the real component of $O(x_0, y_0, \theta_0)$ to represent the output responses in V1 geometry.
- However, it is not the unique way: imaginary component, energy value ($\text{Re}^2(O) + \text{Im}^2(O)$) etc. are used as well in the literature.
- To come back to the retinal (image) plane, an inverse procedure should be applied for all $(x_0, y_0) \in M$:

$$I(x_0, y_0) = \int_{[0, \pi)} O(x_0, y_0, \theta_0) d\theta_0$$

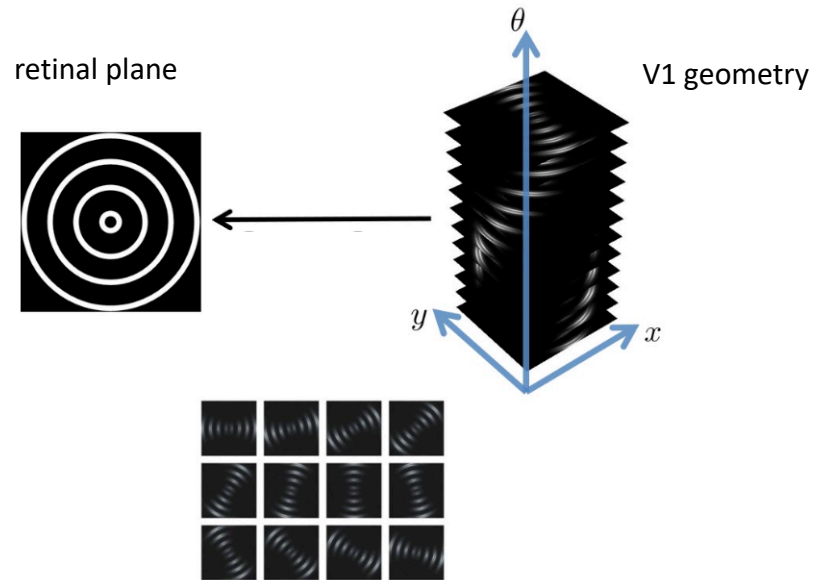
Citti-Sarti-Petitot model of the functional architecture



$$O(x_0, y_0, \theta_0) = \int_M I(x, y) \Psi_{(x_0, y_0, \theta_0)}(x, y) dx dy$$

$$I(x_0, y_0) = \int_{[0, \pi)} O(x_0, y_0, \theta_0) d\theta_0$$

Citti-Sarti-Petitot model of the functional architecture



$$O(x_0, y_0, \theta_0) = \int_M I(x, y) \Psi_{(x_0, y_0, \theta_0)}(x, y) dx dy$$

$$I(x_0, y_0) = \int_{[0, \pi)} O(x_0, y_0, \theta_0) d\theta_0$$

The end